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PyAutoFEP: An Automated Free Energy Perturbation Workflow for GROMACS Integrating Enhanced Sampling Methods

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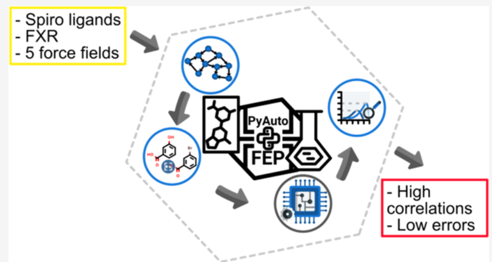
ABSTRACT: Free energy perturbation (FEP) calculations are now routinely used in drug discovery to estimate the relative FEB (RFEB) of small molecules to a biomolecular target of interest. Using enhanced sampling can improve the correlation between predictions and experimental data, especially in systems with conformational changes. Due to the large number of perturbations required in drug discovery campaigns, the manual setup of FEP calculations is no longer viable. Here, we introduce PyAutoFEP, a flexible and open-source tool to aid the setup of RFEB FEP. PyAutoFEP is written in Python3, and automates the generation of perturbation maps, dual topologies, system building and molecular dynamics (MD), and analysis. PyAutoFEP supports multiple force fields, incorporates replica exchange with solute tempering (REST) and replica exchange with solute scaling (REST2) enhanced sampling methods, and allows flexible λ values along perturbation windows. To validate PyAutoFEP, it was applied to a set of 14 Farnesoid X receptor ligands, a system included in the drug design data resource grand challenge 2. An 88% mean correct sign prediction was achieved, and 75% of the predictions had an error below 1.5 kcal/mol. Results using Amber03/GAFF, CHARMM36m/CGenFF, and OPLS-AA/M/LigParGen had Pearson's r values of 0.71 ± 0.13, 0.30 ± 0.27, and 0.66 ± 0.20, respectively. The Amber03/GAFF and OPLS-AA/M/LigParGen results were on par with the top grand challenge 2 submissions. Applying REST2 improved the results using CHARMM36m/CGenFF (Pearson's r = 0.43 ± 0.21) but had little impact on the other force fields. CHARMM36-YF and CHARMM36-WYF modifications did not yield improved predictions compared to CHARMM36m. Finally, we estimated the probability of finding a molecule 1 pK_i better than a lead when using PyAutoFEP to screen 10 or 100 analogues. The probabilities, when compared to random sampling, increased up to sevenfold when 100 molecules were to be screened, suggesting that PyAutoFEP would likely be useful for lead optimization. PyAutoFEP is available on GitHub at https://github.com/lmmpf/PyAutoFEP.



■ INTRODUCTION

Computer-aided drug discovery has become of utmost relevance both in academia and industry.[1,2] In particular, the accurate prediction of the free energy of binding of small molecules to biomolecular receptors is a top priority. Toward this goal, recent studies have demonstrated the precision of FEP calculations and related methods, such as thermodynamic integration, in large-scale retrospective studies,[3,4] and successful application in prospective real drug discovery scenarios.[4−6] Modified protocols have been designed to reach higher accuracy in challenging systems, such as transformations including scaffold hopping,[9] GPCRs,[7,8] and charge perturbations,[10] although the latter may still be affected by artifacts.[11] Charge-changing perturbations are still a matter of active research.[12] Within the past decade, FEP has shifted from a technique suitable for model systems and small transformations[13−15] to one of the most accurate and widely applied methods for estimating relative FEB (RFEB) in drug discovery.[16−18] Applying FEP to lead optimization often requires dozens or even hundreds of perturbations, rendering manual setup unfeasible. Although automated FEP workflows have been previously reported,[3,19−21] to the best of our knowledge, the commercial application, FEP+,[3] is the only one that integrates enhanced sampling. To address the lack of open-source FEP automation software for integrating enhanced sampling protocols, we have developed PyAutoFEP for the automated setup and analysis of FEP calculations with GROMACS.

PyAutoFEP automates the setup, execution, and analysis of RFEB FEP calculations for a series of ligands of a macromolecular target by generating perturbation maps and topologies, running input files, optionally incorporating enhanced sampling methods and flexible λ values, and $\Delta\Delta G$ estimation. Perturbation maps specify the network of transformations, which are used to set up RFEB FEP by simulating the nonphysical transformation of molecule A to B in complex

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and in water (Figure 1). From the complex and water leg simulations, $\Delta\Delta G_{A\rightarrow B}$ can be estimated. Enhanced sampling

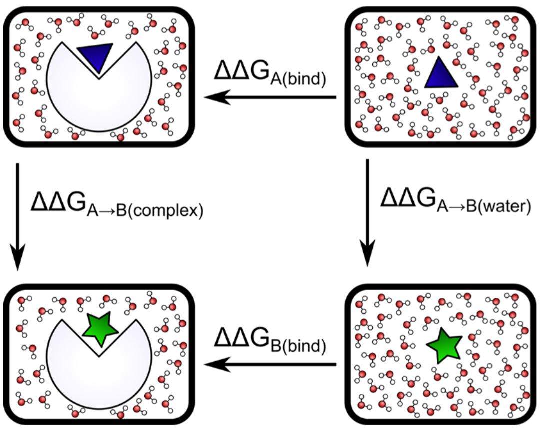

Figure 1. Thermodynamic cycle used in RFEB FEP calculations. The RFEB between molecules A (blue triangle) and B (green star) can be calculated either by simulating the binding or unbinding events of A and B (horizontal arrows) or from the nonphysical transformation of A to B in complex and water (vertical arrows). Due to the computational challenges involved in directly estimating binding events, most approaches calculate RFEB based on the nonphysical transformation of related compounds.

methods were integrated into PyAutoFEP because obtaining adequate sampling with FEP for RFEB estimation is a challenge, especially when protein or ligand rearrangements occur or a large number of atoms are perturbed.[22] Many enhanced sampling algorithms, such as replica exchange molecular dynamics (λ-REMD),[23,24] replica exchange with solute tempering (REST),[25] and replica exchange with solute scaling (REST2),[26] have been shown to improve the convergence of FEP calculations.[27] PyAutoFEP can optionally apply REST and REST2 scaling to solute atoms, and Hamiltonian replica exchange (HREX) can also be used. HREX uses multiple replicas simulated in parallel using different Hamiltonians, so that the coordinates are periodically swapped between replicas. Applying replica exchange can lead to enhanced mixing of the phase space traversed by the system, effectively reducing the length of the simulation and thus the computational effort required to obtain a converged estimate of a property of interest.[24] The efficiency of phase space mixing is improved when the replica exchange connects the end states.[28] A replica exchange connecting the end states, however, requires the use of a unified λ protocol, so that charges and van der Waals interactions are scaled simultaneously, or the use of different λ multipliers for each end state. The former can lead to problems, such as numerical instabilities,[29] which can be circumvented by the use of softcore interactions for both Coulomb and van der Waals potentials,[30,31] while the latter is not supported by GROMACS[32] and AMBER[33] but has been implemented in GROMOS[34] and Desmond.[35] In PyAutoFEP, we implemented a flexible λ protocol, such that charge and van der Waals terms can be scaled independently, thus allowing the replica exchange to connect the end states and maximize mixing.

Here, PyAutoFEP was applied to the Farnesoid X receptor (FXR), comparing five different force fields and the effect of REST2 enhanced sampling. The FXR is a nuclear bile acid receptor involved in the regulation of bile secretion, cholesterol homeostasis, and control of glycemic levels[36,37] and is a drug target for treating metabolic disorders associated with obesity and type 2 diabetes.[38] Several crystallographic structures of the FXR with different ligands are available along with structure–activity relationship data, making it an ideal system for free energy benchmark studies. In 2016, the drug design data resource group (D3R) presented the community challenge grand challenge 2 (GC2) including the blind prediction of binding modes and ligand ranking for two series of FXR ligands.[39] We selected one of them, the spiro series, as an implementation test for PyAutoFEP. The potency of the series spans 3 orders of magnitude, with K_i values ranging from 42 μM (ΔG = −5.97 kcal/mol) to 28 nM (ΔG = −10.30 kcal/mol), and an experimental binding mode is available (Figure 2). The results obtained using PyAutoFEP are on par with top GC2 submissions, indicating that PyAutoFEP can yield accurate RFEB predictions.

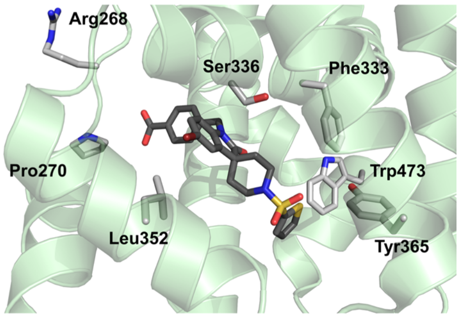

Figure 2. FXR in complex with **FXR_10** (PDB ID 5Q17[39]). The FXR is represented in cartoon style (light green), with binding site residues colored in light gray and **FXR_10** carbon atoms colored in dark gray.

■ MATERIALS AND METHODS

PyAutoFEP Pipeline. PyAutoFEP was written as a series of Python3 commandline scripts, allowing for fine control of the FEP setup process. The program aims to be as user-friendly as possible, while allowing for flexibility in use. Reasonable defaults are provided, allowing the setup of a FEP campaign with minimal user intervention. PyAutoFEP tries to be smart regarding input data (e.g., detecting file formats) and tolerates some level of imperfection in the inputs (e.g., molecular files not fully adhering to standards), while being as clear as possible when a fatal error occurs. A user manual containing a thorough description of the algorithms used by PyAutoFEP is available on GitHub (https://github.com/lmmpf/PyAutoFEP/blob/master/docs/Manual.pdf).

PyAutoFEP comprises a perturbation map generator, a module to prepare dual topologies and prepare and run the simulations, and a result analyzer (Figures 3 and S1). To start, PyAutoFEP reads user-provided SMILES, mol, or mol2 files of the input molecules, one molecule per file, using RDKit[40] and uses the maximum common substructure (MCS) to determine the common core between all pairs. Users can choose between the 3D-guided or graph-based MCS. The 3D-guided MCS is designed to be used when asymmetric centers are inverted in the perturbation, while the graph-based MCS works in all other cases. The 3D-guided MCS iteratively calculates a graph-based MCS between the input molecules excluding chiral centers being perturbed and uses flexible alignment to obtain conformations with maximum overlap, expanding the matching ?

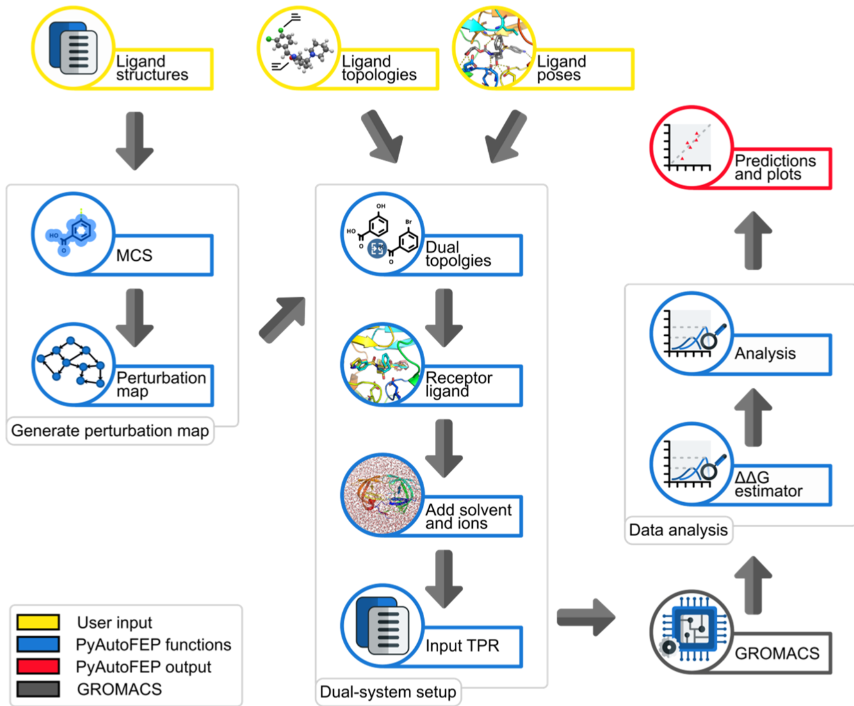

Figure 3. Overview of the PyAutoFEP workflow. The yellow, blue, and red cells correspond to user inputs, PyAutoFEP internals, and outputs, respectively. Gray squares highlight the three main parts of the workflow.

region each step. The 3D-guided MCS algorithm can handle complex molecules, such as carbohydrates and macrocycles. Optionally, users can provide a custom MCS for any number of pairs of molecules. PyAutoFEP represents perturbation maps as directed networkx, a Python3 graph library[41] graphs and generates them by simultaneously minimizing the total number of perturbations and distances between molecule pairs (i.e., the number of perturbations in the shortest path connecting the molecules), using a cost function inspired by Lead Optimization Mapper[42] (Supporting Information 1). The PyAutoFEP map generator uses a modified ant colony optimization (ACO) algorithm to obtain a graph minimizing the cost function, so that the optimization is efficient. The ACO implementation is parallelized and can complete several hundred optimization cycles (typical for a perturbation map with 20−30 nodes) in under a minute on desktop hardware (intel i7-2600 CPU, 16 GB RAM). Currently, the algorithm requires pairwise MCS calculation, which is the slowest step when generating a perturbation map. The pairwise MCS is parallelized, reducing computational time when multiple cores are available. Furthermore, MCS data are saved for further use in the dual-system setup (see below).

The second part of PyAutoFEP generates dual topologies, builds the systems, and prepares GROMACS-compatible inputs. Initially, GROMACS-compatible ligand topologies provided by the user are read. The topology parser works with Amber, OPLS, and CHARMM force fields in the GROMACS format. For each λ state, PyAutoFEP generates dual topologies of the ligands according to the perturbation map, with charge and van der Waals parameters for the perturbed atoms, allowing for complete decoupling of charge and van der Waals λ scaling between end states (eq 1, and Figure S1), so that flexible perturbation schemes can be used.

$$H = H_0 + \lambda_{\mathrm{coul,A}} H_{\mathrm{coul,A}} + \lambda_{\mathrm{VdW,A}} H_{\mathrm{VdW,A}} + \lambda_{\mathrm{coul,B}} H_{\mathrm{coul,B}} + \lambda_{\mathrm{VdW,B}} H_{\mathrm{VdW,B}} \quad (1)$$

where H is the total Hamiltonian of the system, H_0 is the Hamiltonian of the unperturbed region, $H_{\mathrm{coul,A}}$ and $H_{\mathrm{coul,B}}$ are the electrostatic contributions of the particles present only in states A and B, respectively, and $H_{\mathrm{VdW,A}}$ and $H_{\mathrm{VdW,B}}$ are the van der Waals contributions of the particles present only in states A and B, respectively.

Next, PyAutoFEP reads a receptor structure and ligand poses provided by the user. Ligand poses can be read from mol2, pdbqt (e.g., from a docking result), or pdb (e.g., from a crystal structure or MD frame) files, or generated using the built-in core-constrained ligand alignment function. The core-constrained alignment superimposes ligands to a reference pose using a common core and samples conformations of the flexible groups, if any, optimizing the overlap volume. PyAutoFEP then combines the ligand poses and the dual topologies to prepare a receptor−dual-ligand complex for each λ state. A solvation box and counterions are added using GROMACS tools. After the solvated system is built, REST,[25] as described by Terakawa,[43] or REST2[26] can be optionally applied to the solute. Running HREX MD requires a GROMACS compilation patched with a suitable Plumed[44] version. Finally, PyAutoFEP prepares scripts for running the MD on clusters using SLURM/Torque or PBS job manager or computers without job managers, allowing for easy migration between different types of computer infrastructure. System building and input preparation take around half a minute per perturbation on a desktop computer (intel i7-2600 CPU, 16 GB RAM).

After MD completion, PyAutoFEP can use either the Bennett Acceptance Ratio (BAR) or Multistep BAR,[45] implemented in alchemlyb,[46] to estimate the ΔΔG between

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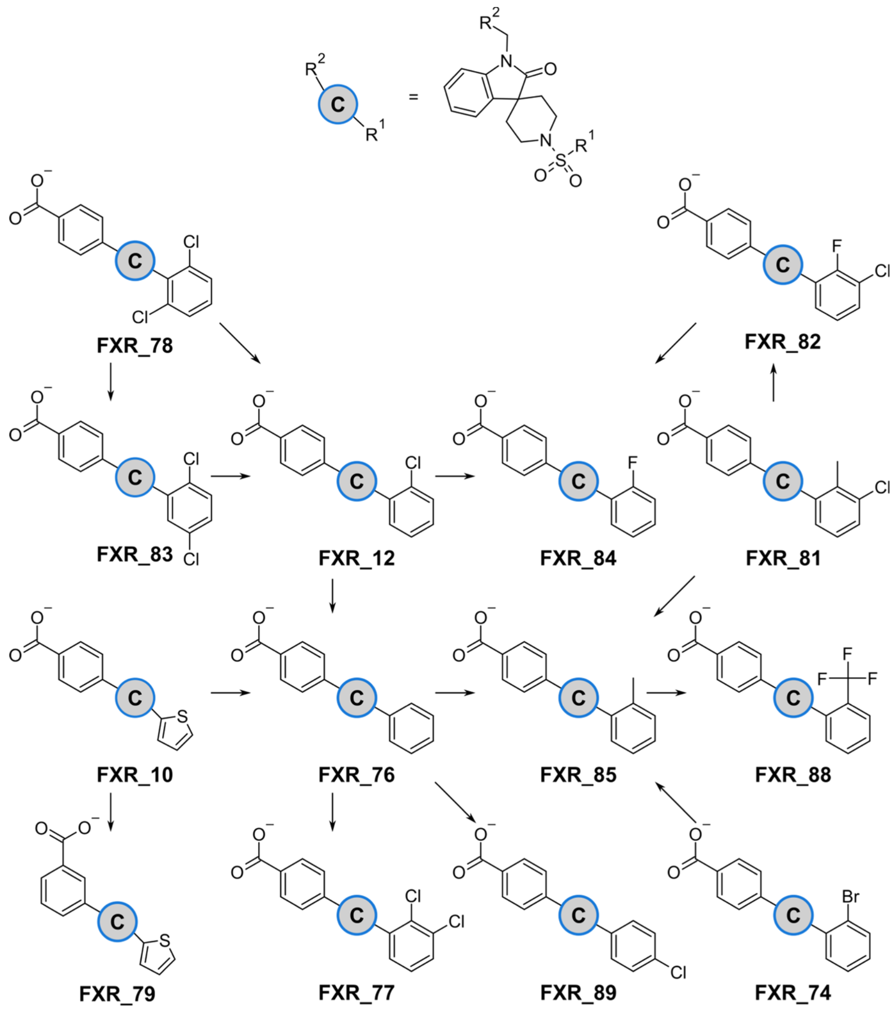

Figure 4. Perturbation map used to calculate the RFEB and molecular structures of the spiro series of FXR binders.

states and associated error. The analysis module uses alchemlyb-analysis[47] and can generate plots for assessing λ-state overlap (overlap matrix, Figure S2A), convergence ($\Delta\Delta G$ *vs* simulation time, Figure S2B), and replica exchange (empirical transition matrix and replica sampling per Hamiltonian, Figure S2C,D, respectively). Furthermore, PyAutoFEP uses GROMACS analysis tools to calculate protein and ligand RMSD, RMSF, distance, and SASA.

FEP Calculations on FXR Spiro Ligands. FXR crystal structures, binding affinity data, and molecular structures were obtained from the D3R's GC2.[39] A total of 14 out of 18 spiro ligands from free energy set 2, which are predicted to be monoanions under assay conditions using the ChemAxon MarvinSketch version 20.3 pK_a plugin, were used (Figure 4). Protein protonation at pH 7.0 was predicted using PropKa.[48] His317 was predicted to be positively charged and participate in a hydrogen bond network with Glu318 and Asp754. All other residues were predicted to be in their standard protonation states. Histidines 298, 348, 426, 449, 450, 451, and 746 were singly protonated on the δ nitrogen, while 258 and 433 were singly protonated on the ε nitrogen. The starting position of the ligands was obtained by core-constrained flexible alignment to the reference crystal structure ligand (**FXR_10**; PDB ID 5Q17,[39] Figure 2) from the GC2 dataset. The alignment was done using the graph-based MCS as the source of constraints; 500 trial poses were generated and ranked using the Shape Tanimoto algorithm,[40] and the structure most similar to the reference was selected. The perturbation map was generated with 500 runs and default scoring parameters (Figure 4).

prepare the protein

MD was performed in triplicate for each perturbation using GROMACS 2019.2[32] patched with Plumed 2.5.2.[44] Standard GROMACS tools were used for solvent and ion addition, as previously described.[49] The minimum distance from the solute to box faces was 1.0 nm, and Na^+ and Cl^- ions were added to achieve charge neutrality and a concentration of 0.15 mM (for some water systems using CHARMM36 variants, 1.5 nm minimum distance to the box faces was used, see Supporting Information 1 and Table S1). The systems were minimized and then equilibrated using *NVE*, *NVT*, and *NPT* ensembles successively for 0.2 ns each. For FEP, systems were sampled using the unrestrained *NPT* ensemble for 10 ns at 1.0 atm and 298.15 K. The Verlet cutoff scheme was used, PME order was 4, and Fourier spacing was 0.12 nm. Bonds connecting hydrogens to heavy atoms were constrained using LINCS.[50] Simulations were run in the forward direction only (i.e., according to Figure 4), using 12 λ windows (Table S2) for all perturbations and also using 24 λ windows for **FXR_10** → **FXR_79** (Table S3). Pairwise RFEB was converted to RFEB with respect to **FXR_10** by summing the $\Delta\Delta G$ of perturbations connecting each molecule to **FXR_10** using the shortest path in the perturbation map (i.e., the shortest

10ns in total

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path, given by the number of perturbations connecting each molecule to **FXR_10** in the graph). Where REST2 was used, a scaling factor of 0.5[26] on the central window was applied to all atoms of the ligand, with a linear scaling towards the end-states, and Hamiltonian replica exchange was attempted every 1.0 ps. The scaling factor reduces the potential energy barriers on the ligand, allowing faster sampling of the phase space and enhanced replica exchange acceptance.[26]

For FEP using Amber03/GAFF, ligands were parameterized using AcPype[51] with the generalized amber force field (GAFF) and AM1-BCC charges;[52] Amber03[53] was used to model the protein, and TIP3P was used as the water model. For FEP using CHARMM36 variants, ligands were parametrized using CGenFF[54] and converted to a GROMACS-compatible format. CHARMM36m,[55,56] CHARMM36-YF,[57] and CHARMM36-WYF[58] were used to model the protein, and CHARMM-modified TIP3P[59] was used as the water model.[56] For FEP using OPLS-AA/M/LigParGen, ligands were parameterized using LigParGen;[60] OPLS-AA/M[61] was used to model the protein, and TIP4P[62] was used as the water model. For Amber03/GAFF and OPLS-AA/M/LigParGen, Coulomb and van der Waals short-range cutoffs were 1.0 nm and dispersion correction was employed, while for all CHARMM36 variants, no dispersion correction was applied,[56] and Coulomb and van der Waals short-range cutoffs were 1.2 nm, with force switching at 1.0 nm for van der Waals interactions.

Statistical Analysis. Triplicate datasets were analyzed by averaging each predicted $\Delta\Delta G$ and calculating the correlation statistics using the mean. Centered root mean square error (RMSEc) (eq 2), Kendall's τ (from here on referred to as τ), Spearman's rank correlation coefficient, Pearson correlation coefficient, and their associated errors were obtained using D3R's scripts (https://github.com/drugdata/D3R_grandchallenge_evaluator) using 10,000 bootstrap rounds. The statistics are reported along with the respective errors. Standard deviation is reported along with means between replicas. Overlap matrices were obtained using pymbar[45] implementation through PyAutoFEP.

$$\mathrm{RMSE_c} = \sqrt{\frac{\sum((\hat{x}_i - x_i) - \overline{\varepsilon})^2}{N}} \qquad \overline{\varepsilon} = \frac{\sum(\hat{x}_i - x_i)}{N} \tag{2}$$

where N is the number of predictions, $\hat{x}_i$ is the predicted $\Delta\Delta G$, and x_i is the experimental $\Delta\Delta G$ for a particular molecule.

Statistical hypothesis testing was done using SciPy. One-sample Student's t-test was used for testing if the mean of a population deviates from a theoretical mean. Levene's test was used to test the equality of the variances between groups, and the median was used as the center for the distribution.[63] Two-sample Kolmogorov–Smirnov[64] was used to test if underlying distributions of samples were statistically different. A significance level of α = 0.01 was used. When multiple hypotheses were tested, Bonferroni correction[65] was applied.

RESULTS AND DISCUSSION

PyAutoFEP Results Were Comparable to Top GC2 Performers. Using PyAutoFEP, FEP calculations were performed using five force fields with and without REST2. A relevant criterion for lead optimization is the ability to correctly predict whether a modification in the ligand structure leads to an increase ($\Delta\Delta G < 0.0$) or a decrease ($\Delta\Delta G > 0.0$) in affinity, allowing prioritization of compounds for synthesis.[4] Independent of force fields or the use of enhanced sampling, the correct sign was recovered for 10−13 (77−100%) of the predictions (Tables 1 and 2, Supporting Information 2), with a mean of 11 ± 1 (88%). In the absence (presence) of REST2, most of the predictions were within 2 kcal/mol of their experimental value: 77% (85%) for Amber03/GAFF, 85% (92%) for CHARMM36m/CGenFF, and 85% (100%) for OPLS-AA/M/LigParGen (Figure 5, Table 1). Correlations between predicted and calculated $\Delta\Delta G$ values ranged between weak and strong and were dependent on the force field, sampling regime, and number of λ windows (Table 2). Using the Pearson correlation metric, Amber03/GAFF and OPLS-AA/M/LigParGen had the highest agreement (Pearson's r = 0.71 ± 0.13 and 0.66 ± 0.20, respectively), followed by CHARMM36m/CGenFF (0.28 ± 0.27). The CHARMM36m/CGenFF predictions gave a rather flat correlation (r = 0.08 without REST2 and r = 0.14 with REST2) and may have suffered from a systematic bias, especially when REST2 was not applied (mean error = 1.01 ± 1.24 kcal/mol; one-sample Student's t-test for mean ≠ 0.0 p-val = 0.0092). Therefore, in the absence of REST2, CHARMM36m/CGenFF overestimated the predictions by an average of 1 kcal/mol with respect to FXR_10 (Table 2). When REST2 was applied, CHARMM36m/CGenFF was able to better reproduce the potency trends in the dataset (Spearman's r = 0.51 ± 0.23 with REST2 *vs* Spearman's r = 0.19 ± 0.31 without REST2, which measures the rank relationship between experimental and predicted values, irrespective of linearity), but struggled to capture the magnitudes of RFEB.

Our results for the FXR spiro ligands obtained with PyAutoFEP were on par with leading Free Energy-based GC2 submissions (Supporting Information 1, Table S4). The first-place submission, xk67c, combined umbrella sampling and Jarzynski nonequilibrium pulling along a dissociation coordinate, yielding the highest Kendall's τ (τ = 0.62 ± 0.12) and lowest error (RMSEc = 0.94 ± 0.15 kcal/mol). The second-place submission, 81n55, used FEP+ and reported τ = 0.55 ± 0.12 and RMSEc = 1.31 ± 0.20 kcal/mol. The third-place submission, jzrt5, used Amber14SB/GAFF and a charging scaling method and reported τ = 0.37 ± 0.14 and RMSEc 1.45 ± 0.23 kcal/mol. It is noteworthy that the original submissions predicted RFEB for the entire dataset, while we did so for 14 out of the 18 ligands, which do not involve charge modification. Based on τ, our results using Amber03/GAFF and no REST2 would have ranked second with τ = 0.59 ± 0.12, and OPLS-AA/M/LigParGen with no REST2 would have ranked third with τ = 0.50 ± 0.19. Based on RMSEc, our predictions using OPLS-AA/M/LigParGen and no REST2 would have ranked first, with a value of 0.86 ± 0.17 kcal/mol, while Amber03/GAFF with no REST2 and CHARMM36m/CGenFF with REST2 would have ranked second, with values of 1.03 ± 0.17 and 1.03 ± 0.19 kcal/mol, respectively (Table S4). Our predictions using CHARMM36m/CGenFF without REST2 had a lower RMSEc (1.20 ± 0.19 kcal/mol) than submission izhdb (RMSEc = 1.61 ± 0.28 kcal/mol; τ = 0.28 ± 0.17), which used CHARMM36/CGenFF but employed multi-site λ dynamics instead of FEP and CHARMM[66] instead of GROMACS (Table S4). Although PyAutoFEP allows considerable flexibility in the system setup and simulation parameters, the FXR predictions performed here did not employ specialized methods and yielded good results,

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Table 1. Mean Predicted $\Delta\Delta G$ and Standard Deviations with Respect to FXR_10[a]

molecule	$\Delta\Delta G_{expt}$ (kcal/mol)	Amber03/GAFF/AM1-BCC		CHARMM36m/CGenFF		CHARMM36-YF/CGenFF	CHARMM36-WYF/CGenFF	OPLS-AA/M/LigParGen	
		without REST2	with REST2	without REST2	with REST2	without REST2	without REST2	without REST2	with REST2
FXR_12	**−2.7**	−2.4 ± 0.3	−3.0 ± 0.6	−1.7 ± 0.5	−1.2 ± 0.6	−1.6 ± 0.8	−1.0 ± 0.3	−0.6 ± 0.7	−0.7 ± 0.7
FXR_74	**−1.3**	−1.8 ± 1.1	−2.9 ± 1.0	−2.7 ± 0.8	−1.1 ± 0.6	−1.0 ± 1.0	−1.8 ± 0.6	−0.6 ± 0.9	−1.2 ± 0.2
FXR_76	**1.2**	0.0 ± 0.3	−0.7 ± 0.3	−1.6 ± 0.6	−1.3 ± 0.5	−1.4 ± 0.4	−1.3 ± 0.3	0.5 ± 0.4	0.2 ± 0.1
FXR_77	**−1.8**	−3.3 ± 0.3	−4.7 ± 0.7	−2.3 ± 0.9	−2.3 ± 0.8	−2.0 ± 0.6	−1.6 ± 0.4	−2.7 ± 1.3	−2.7 ± 1.1
FXR_78	**−3.1**	−3.6 ± 0.4	−3.4 ± 0.5	−2.1 ± 0.9	−2.2 ± 0.8	−0.9 ± 0.8	0.5 ± 1.7	−1.9 ± 0.9	−1.7 ± 1.2
FXR_79 (24 λ)	**−0.2**	1.3 ± 1.0	2.4 ± 0.9	−2.6 ± 0.9	−1.8 ± 2.0	−1.9 ± 2.3	−4.2 ± 1.3	−0.8 ± 2.3	4.5 ± 6.4
FXR_79 (12 λ)[b]	**−0.2**	3.4 ± 2.7	6.9 ± 5.1	0.1 ± 16.5	−0.3 ± 5.8	−11.6 ± 14.3	−2.5 ± 15.5	2.0 ± 2.9	−3.6 ± 7.3
FXR_81	**−0.4**	−2.2 ± 0.9	−3.3 ± 1.0	−3.2 ± 0.8	−2.2 ± 0.5	−1.5 ± 1.0	−2.8 ± 0.6	−1.8 ± 1.1	−1.6 ± 1.3
FXR_82	**−2.0**	−2.1 ± 1.1	−2.6 ± 0.6	−2.3 ± 1.2	−2.1 ± 0.9	−1.7 ± 0.9	−1.2 ± 0.1	−1.5 ± 0.9	−2.1 ± 0.5
FXR_83	**−1.7**	−2.9 ± 0.7	−2.8 ± 0.4	−2.4 ± 0.3	−2.1 ± 0.8	−3.1 ± 0.6	−1.8 ± 0.3	−1.4 ± 1.6	−2.1 ± 0.3
FXR_84	**−0.1**	−0.5 ± 0.9	−0.7 ± 0.4	−1.1 ± 0.8	−0.9 ± 0.5	−1.0 ± 0.3	−0.2 ± 0.4	0.2 ± 0.5	−0.2 ± 0.3
FXR_85	**−1.7**	−0.6 ± 0.8	−1.6 ± 0.9	−2.9 ± 0.8	−1.4 ± 0.4	−0.9 ± 1.0	−1.7 ± 0.8	−1.5 ± 0.7	−1.9 ± 0.5
FXR_88	**−1.4**	−1.8 ± 1.1	−3.2 ± 0.7	−3.6 ± 1.1	−1.8 ± 0.7	−1.0 ± 0.4	−2.2 ± 1.4	−1.6 ± 0.7	−0.8 ± 1.1
FXR_89	**−1.2**	0.5 ± 0.2	−0.9 ± 0.8	−2.3 ± 0.4	−1.2 ± 0.5	−2.0 ± 0.5	−1.7 ± 0.3	−0.9 ± 0.7	−1.4 ± 0.2

[a]Predictions using 12 λ windows shown here for context. The data using 24 λ for **FXR_79** are reported throughout the paper. [b]All values are given in kcal/mol.

suggesting that utilization of standardized protocols does not jeopardize calculation accuracy.

The superior performance of Amber03/GAFF and OPLS-AA/M/LigParGen compared to that of CHARMM36m/CGenFF was in line with previously reported data[20,67] and may be at least partially attributed to the parameterization bias. The possibility that large fluctuations of the ligand poses negatively impacted the performance of CHARMM36m/CGenFF was evaluated. Analysis of the stability of ligand poses during the MD showed that while a slightly increased ligand RMSD was observed for CHARMM36m/CGenFF in comparison to that of Amber03/GAFF and OPLS-AA-M/LigParGen, the final poses were similar to the crystal structure (Figure S3). Then, characteristics of the CHARMM36m/CGenFF parameters and previous reports of its performance in comparison with other force fields were analyzed. In contrast to Amber03/GAFF and OPLS-AA/M/LigParGen, CHARMM36m/CGenFF allows for the treatment of halogen bonds using a charged lone pair added to aromatic halogens,[54] which, presumably, should have led to better results as the compounds under investigation bear aryl halogens. Other studies have compared CGenFF with other small-molecule force fields. Jespers et al. compared OPLS-AA/M, Amber ff14SB/GAFF, and CHARMM36/CGenFF for FEP calculations on a series of cyclin-dependent kinase 2 using the program Q and the automation tool QLigFEP.[20] MUE with CHARMM36/CGenFF was 1.54 kcal/mol, while MUE with Amber ff14SB/GAFF was 1.11 kcal/mol. CHARMM36/CGenFF also performed worse than OPLS-AA/M/OPLS/CM5 and OPLS-AA/M/OPLS/CM1A but better than OPLS/AA//OPLS/CM1A, in a small study of two flavin nucleotides binding to flavodoxin.[67] CGenFF was parameterized to reproduce QM-level MP2/6-31G(d) Merz−Kollman charges[54] and then refined to reproduce distance and enthalpy of the interaction with a water molecule. It is possible that the parameterization better captures the interactions of small molecules with water and pure liquids, which were extensively validated during CGenFF development, rather than the interactions of small molecules with biomolecular receptors. In line with this hypothesis, Kashefolgheta et al.[68] observed that solvation free energies obtained using CGenFF for *N,N*-dimethylacetamide against 22 solvents had a relatively high error (RMSE = 2.75 kcal/mol) compared to several other solutes. A large solvation free energy error for an amide may suggest problematic amidic interaction parameters, which could also be shared by main chain atoms in proteins.

Two variants of CHARMM36, CHARMM36-YF and CHARMM36-WYF, did not yield improved predictions over the CHARMM36m results. CHARMM36-YF is a modified version of CHARMM36m where phenylalanine and tyrosine Lennard-Jones (LJ) parameters were refitted to reproduce a reference cation−π interaction potential energy surface at the *ab initio* level (SAPT2+/aug-cc-pVDZ).[57] CHARMM36-WYF extends this same idea to tryptophans.[58] Although no cation−π interaction with any of the ligands is possible, we were interested in testing the effects of the modified parameters on the LJ interactions between the ligand and aromatic residues in the binding site, such as Phe333, Tyr365, and Trp473 (Figure 1). Because no hydrogen bond or ion−ion interactions between the ligand and FXR were evident from the crystal structure, LJ parameters were expected to have a high impact on binding affinity prediction. Also, a better representation of the cation−π interaction in the binding site between Arg335

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Table 2. Statistics Based on the Mean ΔΔGs

Force field	REST2	Kendall's τ[a]	Spearman's r[a]	Pearson's r[a]	RMSEc (kcal/mol)[a]	mean error[b] ± SD (kcal/mol) (p-val[c])	MAE[d] ± SD (kcal/mol)	correct sign[e] (%)
Amber03/GAFF	**No**	0.59 ± 0.12	0.77 ± 0.13	0.71 ± 0.13	1.03 ± 0.17	0.20 ± 1.07 (p-val = 0.4948)	0.86 ± 0.62	77
	Yes	0.49 ± 0.15	0.69 ± 0.17	0.63 ± 0.12	1.37 ± 0.12	0.77 ± 1.42 (p-val = 0.0632)	1.21 ± 1.05	85
CHARMM36M/CGenFF	**No**	0.07 ± 0.23	0.19 ± 0.31	0.30 ± 0.27	1.20 ± 0.19	1.01 ± 1.24 (p-val = 0.0092)	1.31 ± 0.90	85
	Yes	0.38 ± 0.19	0.51 ± 0.23	0.43 ± 0.21	1.03 ± 0.19	0.36 ± 1.06 (p-val = 0.2294)	0.78 ± 0.78	92
CHARMM36-YF/CGenFF	**No**	0.05 ± 0.20	0.13 ± 0.27	0.16 ± 0.23	1.23 ± 0.22	0.24 ± 1.27 (p-val = 0.4975)	0.99 ± 0.78	92
CHARMM36-WYF/CGenFF	**No**	−0.27 ± 0.25	−0.21 ± 0.33	−0.24 ± 0.31	1.79 ± 0.38	0.32 ± 1.85 (p-val = 0.5220)	1.24 ± 1.37	85
OPLS-AA/M/LigParGen	**No**	0.50 ± 0.19	0.65 ± 0.22	0.66 ± 0.20	0.86 ± 0.17	−0.14 ± 0.89 (p-val = 0.5598)	0.68 ± 0.57	85
	Yes	0.49 ± 0.15	0.70 ± 0.18	0.52 ± 0.1−4	1.46 ± 0.44	−0.35 ± 1.52 (p-val = 0.4049)	0.91 ± 1.25	100

[a]Statistics and associated errors computed using the D3R evaluator script. [b]Mean of $\Delta\Delta G_{expt} - \Delta\Delta G_{calc}$. SD calculated over the 13 predictions. [c]One-sample Student's t-test for mean ≠ 0.0. [d]SD calculated over the 13 predictions. [e]Proportion of predictions with the same sign as the experimental ΔΔG.

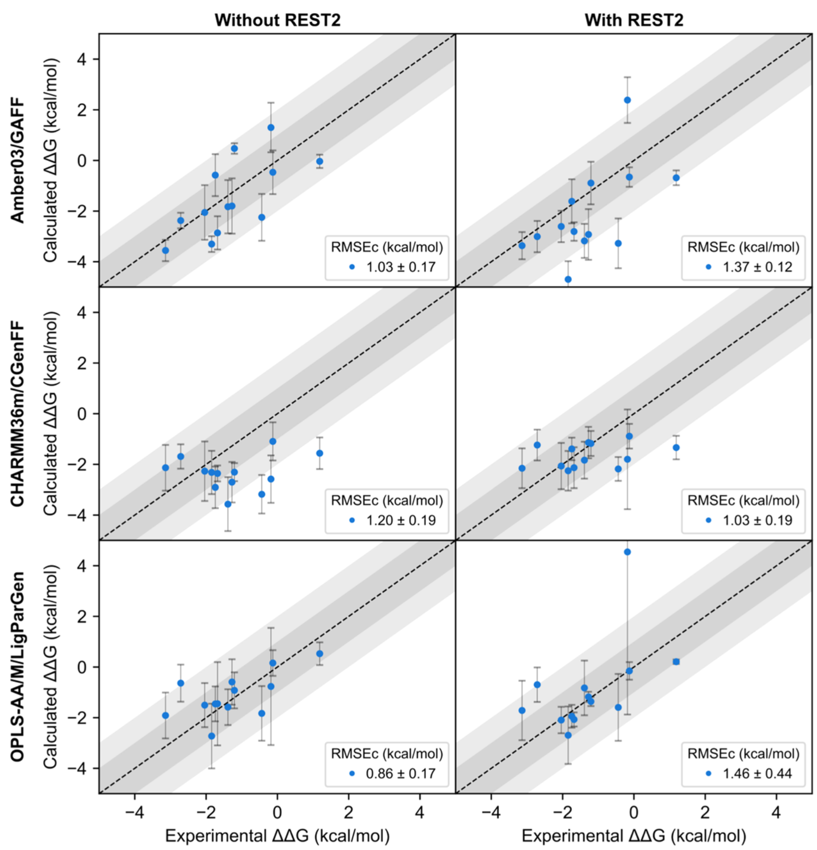

Figure 5. Correlation between experimental and calculated ΔΔG with respect to **FXR_10**. Means and standard deviations are plotted for each prediction, based on three replicates for each transformation. The dashed line is $Y = X$ (i.e., a theoretical perfect correlation). Dark-gray and light-gray regions represent 1 and 2 kcal/mol error ranges, respectively.

and Tyr264 could affect the dynamics of the complex and the estimated RFEB. The CHARMM36-YF/CGenFF results, however, did not improve the correlation with experimental affinities (Pearsons's r = 0.16 ± 0.23), compared to CHARMM36m/CGenFF (Pearsons's r = 0.30 ± 0.27). CHARMM36-WYF/CGenFF yielded inferior predictions than CHARMM36-YF/CGenFF (Pearsons's r = −0.24 ± 0.31). To assess whether the LJ potential between ligands and FXR was substantially affected by the refitted LJ parameters in CHARMM36-YF and CHARMM36-WYF, GROMACS energy groups were used to decompose the potential energy (Figure S4). From the histograms and statistical tests, it is clear that the parameters affected the van der Waals interactions for most of the ligands. The reference ligand **FXR_10** showed the largest shift in mean potential in the series (CHARMM36-YF ΔU = 6.21 kcal/mol, CHARMM36-WYF ΔU = 4.84 kcal/mol). It was also notable that the prediction accuracy of CHARMM36-WYF was considerably lower than that of

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CHARMM36m and CHARMM36-YF, considering that Trp473 is in close contact with the spiro ligands (Figure 2). Given the magnitude of the differences and that statistically significant differences between distributions for most of the ligands were observed, it is possible that the refitted LJ parameters for Phe, Tyr, and Trp worsened the RFEB predictions by affecting the protein−ligand interaction potential.

Two closed cycles are present in the perturbation map used (Figure 4), **FXR_12-FXR_78-FXR_83**, and **FXR_76-FXR_84-FXR_82-FXR_81-FXR_85**, which were used to evaluate cycle closure hysteresis, a measure of the deviation of the $\Delta\Delta G$ sums from 0 in a closed cycle.[69] Absolute errors ranging from 0.6 to 2.5 kcal/mol were obtained for **FXR_12-FXR_78-FXR_83,** and those in the range of 0.2−4.5 kcal/mol were obtained for **FXR_76-FXR_84-FXR_82-FXR_81-FXR_85**(Table S6). Predicted standard deviations of cycle closure errors (σ) were calculated according to Wang et al.,[69] which suggests that for errors above 1 σ, it is highly probable (P = 0.68) that the calculation is not converged and if errors are above 2 σ, it is almost certain (P = 0.95) that the calculation is not converged. Based on the aforementioned criteria, the **FXR_12-FXR_78-FXR_83** cycle using CHARMM36m/CGenFF with REST2 and OPLS-AA/M/LigParGen without REST2 likely did not converge. The **FXR_76-FXR_84-FXR_82-FXR_81-FXR_85** cycle using Amber03/GAFF with and without REST2 and CHARMM36m/CGenFF with REST2 also likely did not converge. Overall, six out of eight conditions yielded good cycle errors for **FXR_12-FXR_78-FXR_83**, and five out of eight conditions yielded good cycle errors for **FXR_76-FXR_84-FXR_82-FXR_81-FXR_85**. Interestingly, the lowest errors for the **FXR_12-FXR_78-FXR_83** (0.6 kcal/mol without REST2 and 1.1 kcal/mol with REST2) and largest errors for the **FXR_76-FXR_84-FXR_82-FXR_81-FXR_85** (4.1 kcal/mol without REST2 and 4.5 kcal/mol with REST2) cycles were observed for Amber03/GAFF, suggesting that it may exhibit a higher hysteresis on some cycles and lower on others. The error values were larger than those reported in Olsson et al.,[70] which reported, for the same series, a mean hysteresis of 1.4 and 0.5 kcal/mol for 3- and 5-membered cycles, respectively. Cycle closure errors deviating from 0.0 kcal/mol have been reported for the FXR[70] and for other systems.[71,72]

For all force fields, we analyzed whether the error distribution suggested systematic bias and the extent of random errors. To assess the systematic bias in the predictions, one-sample Student's t-test was used to evaluate if the mean error would be 0.0 kcal/mol (i.e., whether the error distribution is centered at 0.0 kcal/mol). The hypothesis that the predictions do not suffer from systematic bias for all force fields, with or without REST2 (Table 1), cannot be rejected, except for CHARMM36m/CGenFF. A study applying FEP+ to 330 perturbations for eight targets also observed an unbiased prediction distribution of errors with a roughly Gaussian shape, a standard deviation of 1.14 kcal/mol, and centered at 0.0 kcal/mol.[3] The standard deviations of errors using different force fields without REST2 ranged from 0.89 kcal/mol using OPLS-AA/M/LigPargGen to 1.85 kcal/mol using CHARMM36-WYF/CGenFF, while the standard deviations of errors with REST2 ranged from 1.06 kcal/mol using CHARMM36/CGenFF to 1.52 kcal/mol using OPLS-AA/M/LigParGen. It is worth noting that most of the variance of CHARMM36-WYF/CGenFF predictions, for which the highest standard deviation of errors was observed, came from the **FXR_79** predictions (standard deviation = 1.85 kcal/mol; standard deviation excluding **FXR_79** = 1.57 kcal/mol; Supporting Information 2). The variances between all force fields were statistically equivalent, suggesting a similar performance with regards to error dispersion (Levene's test p-value = 0.8027 for equality of the variances). PyAutoFEP results were also on par with those of large-scale FEP+ studies in an industrial drug discovery context, where spleen Tyr kinase and c-Met kinase datasets had standard deviations of 1.61 and 1.43 kcal/mol, respectively.[18]

Specifically, the **FXR_10** → **FXR_79** perturbation was the most challenging of the series, as large variances and errors were observed, likely because of the charge hopping (Supporting Information 1). The predictions for **FXR_10** → **FXR_79** were vastly improved using 24 λ instead of 12 λ windows, possibly due to better overlap of adjacent states, but more studies are necessary to confirm this hypothesis (Figure S5). FEP runs for 20 ns for the **FXR_10** → **FXR_79** perturbations using 12 λ windows were also tested, but the longer sampling time did not lead to improved predictions (Table S7). In another study, 24 λ windows were employed for ligand pairs involving charge-changing (where the total charge of the system changes between states A and B) and charge-hopping perturbations (where the charge position changes but the total charge is constant between states A and B), resulting in RMSE values of around 1 kcal/mol for the series studied.[10] Errors for **FXR_79** predictions ranged from 1.0 to 4.4 kcal/mol. Despite efforts to improve the treatment of charge-modifying transformations,[10,73−76] these systems remain one of the biggest challenges in FEP simulations.[77]

The prediction trends were reproducible when the replicas were analyzed independently. Mean Pearson's r between replicas using Amber03/GAFF (0.66 ± 0.05), CHARMM36m/CGenFF (0.26 ± 0.24), and OPLS-AA/M/LigParGen (0.51 ± 0.09) was similar to Pearson's r calculated using the mean predicted $\Delta\Delta G$ for these force fields (0.71 ± 0.13, 0.28 ± 0.27, and 0.52 ± 0.14, respectively). The mean standard deviation of all predictions using different force fields and with or without REST2 was 0.80 kcal/mol, suggesting that the replicates had a small dispersion. Also, 84 (81%) out of 104 standard deviations were below 1.0 kcal/mol and 98 (94%) were below 1.5 kcal/mol, suggesting a high precision. Furthermore, out of the 312 individual replica predictions (Supporting Information 2), 53% had an absolute error below 1.0 kcal/mol and 70% had an absolute error below 1.5 kcal/mol. The correlation statistics for individual replicas were also evaluated (Table S5), and the results suggest that the three replicates were reproducible and that single replicas capture most of the trends, although averaging the triplicates did improve the results to some extent.

PyAutoFEP Could Be Useful in Lead Optimization Campaigns. To estimate the utility of PyAutoFEP in drug discovery, a statistical experiment proposed by Shirts et al.[78] was used to assess the effect on the chance of finding better compounds when a screening method is used during lead optimization. The authors used data reported by Hajduk and Sauer[79] on experimental affinities of over 84,000 compounds from lead optimization against 30 targets. The data show that the distribution of changes in affinities of assayed compounds with respect to the reference compound was normally distributed with μ = 0.0 kcal/mol and σ = 1.02 kcal/mol.

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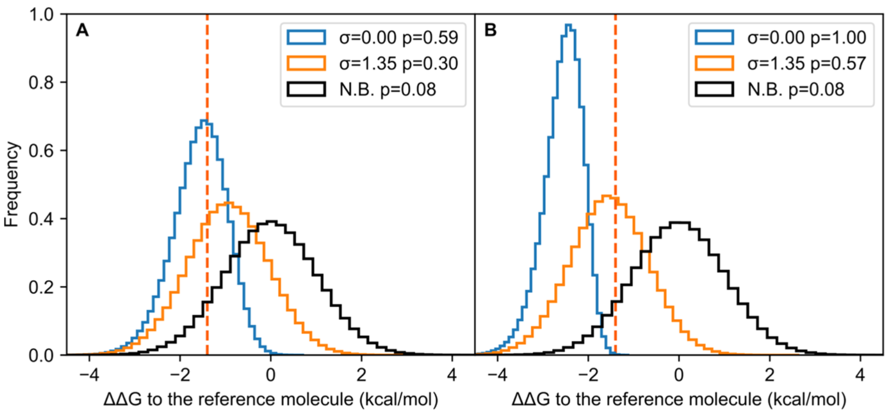

Figure 6. Distribution of the relative affinity of the best-ranked compound chosen from a statistical experiment, employing methods with varied standard deviations. (A) Best-ranked compound selected after screening 10 molecules. (B) Best-ranked compound selected after screening 100 molecules. Molecules are modeled with normal affinity changes (μ = 0.0 kcal/mol and σ = 1.02 kcal/mol) to a theoretical reference compound. The orange dashed vertical line corresponds to −1.4 kcal/mol. The probabilities, p, of finding a compound with a change lower than −1.4 kcal/mol are given. The blue distribution corresponds to molecules selected with a theoretically perfect (no noise, σ = 0.0 kcal/mol) method, orange distribution corresponds to molecules selected with a method with similar standard deviations as our results (σ = 1.35 kcal/mol), and black distribution corresponds to a random, nonbiased selection.

The authors then simulated a lead optimization campaign where the best candidate of N molecules was selected using a method with Gaussian noise with μ = 0.0. The probability of the candidate having an increased potency of at least 1.4 kcal/mol or 1 pK_i unit can be estimated for a given N and σ by running the simulation a large number of times.

The aforementioned statistical experiment was carried out using the standard deviation of the errors observed for the full set of predictions (σ = 1.35 kcal/mol). The simulations were run a million times using N = 10 and N = 100. Random selection would find a compound better by at least 1.4 kcal/mol 8% of the time (see the nonbiased curve in Figure 6). An idealized method with no noise (i.e., a method able to perfectly predict the compound affinity) and using N = 10 would find such a compound 59% of the time because in 41% of the runs, none of the 10 compounds screened were better by more than 1.4 kcal/mol. When screening 10 molecules using a method with the same standard deviation as we observed in this work (σ = 1.35 kcal/mol), the chance of finding better compounds is 30%, representing a fourfold increase in comparison to that with random sampling. Screening 100 compounds would increase the chance of finding a compound better by 1 pK_i to 57%, a sevenfold increase with respect to that of random sampling and almost two times better than with screening 10 molecules with the same method. Thus, our statistical simulation suggests that PyAutoFEP can be useful for tackling lead optimization problems.

CONCLUSIONS

In this work, we introduced PyAutoFEP, a tool for automating FEP calculations for GROMACS, supporting enhanced sampling methods. Although other automation tools exist, PyAutoFEP adds to the community as it is open-source, uses the highly efficient GROMACS code, supports multiple force fields, and integrates enhanced sampling methods. To test PyAutoFEP, we applied it to a series of FXR ligands, which were presented as a community challenge by the D3R group. Amber03/GAFF, OPLS-AA/M/LigParGen, and CHARMM36m/CGenFF force fields were compared with and without REST2 enhanced sampling. Considerable variability in prediction accuracy was obtained depending on the force field, enhanced sampling regime, and number of λ windows, although all setups resulted in a correct sign recovery of at least 77% for the transformations evaluated. Amber03/GAFF and OPLS-AA/M/LigParGen performed better than CHARMM36m/CGenFF. REST2 improved CHARMM36m/CGenFF results but had minor effects on Amber03/GAFF and OPLS-AA/M/LigParGen. Two CHARMM36 variants, CHARMM36-YF and CHARMM36-WYF, did not lead to improved predictions over CHARMM36m. A statistical simulation to assess the usefulness of PyAutoFEP in a drug discovery campaign indicated that the FEP pipeline applied in this work would lead to a significantly increased probability of finding more potent compounds if used to screen molecules in lead optimization. As a whole, our study demonstrates that PyAutoFEP is a useful tool for tackling drug discovery problems.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.jctc.1c00194.

Representation of the dual-topology perturbation model; scoring function for the optimal map generator algorithm; example plots generated by PyAutoFEP analysis scripts; MD systems using 1.5 nm minimum distance to the box faces; perturbations for which a 1.5 nm water leg system using a minimum distance of 1.5 nm to the box faces were used; 12 λ scheme used to represent the perturbations in our study; 24 λ scheme used to represent the perturbations in our study; data for original submissions to drug design data resource GC2 for the spiro series; statistics on individual FEP replicas; ligand RMSD and examples of some of the final ligand conformations; histograms of van de Waals potentials between FXR and each of the spiro ligands; cycle closure hysteresis observed for the two cycles in the perturbation map; comparing 12 and 24 λ windows for a problematic perturbation; mean predicted $\Delta\Delta G$ for **FXR_79** using 10 and 20 ns sampling; overlap matrices for the perturbation **FXR_10** → **FXR_79**; and statistics on

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FEP replicas using Amber03/GAFF without REST2 using 12 or 24 λ windows (PDF)

Spreadsheet containing individual predictions and errors for all molecules and correlation and error statistics for individual replicas using 12 and 24 λ windows for **FXR_10 → FXR_79** (XLSX)

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Notes

The authors declare no competing financial interest.

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